

# SOC INCIDENT REPORT

## Automated Threat Removal via Threat Intelligence Correlation (Wazuh XDR + VirusTotal)

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|------------------------|----------------------------|
| Client / Organization: | Internal Corporate Network |
| Prepared By:           | Md. Kawsar Hossain         |
| Role:                  | SOC Analyst Tier 2         |
| Report Date:           | 1/9/2026                   |
| Report Reference No.:  | IR-2026-09-001             |
| Classification:        | TLP: AMBER (Confidential)  |

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## 1. Executive Summary

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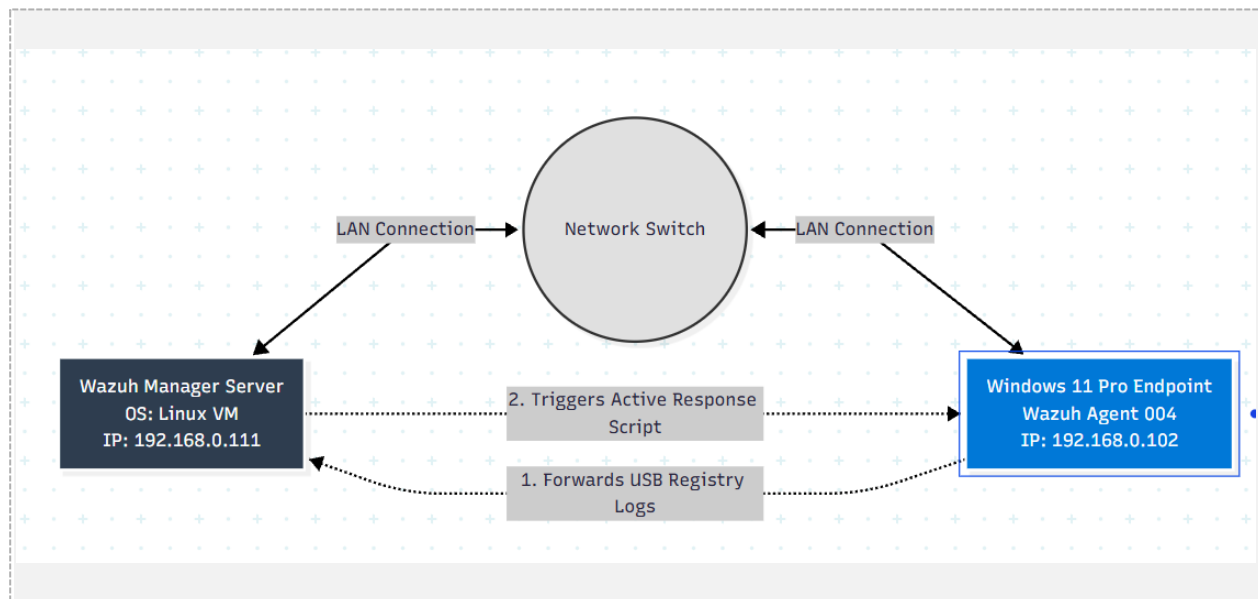
An automated threat containment pipeline successfully neutralized a malware intrusion attempt on a corporate Windows 11 endpoint. Utilizing Wazuh (XDR) and the VirusTotal API, the Security Operations Center (SOC) detected a malicious file drop via real-time File Integrity Monitoring (FIM). Within seconds of detection, the system cross-referenced the file hash with threat intelligence feeds and triggered an Active Response execution, permanently deleting the threat from the host system without requiring manual analyst intervention.

## 2. Attack Summary

### 2.1 Network / Environment Design

**Target Endpoint:** Corporate Workstation / Windows 11 Pro (IP: 192.168.0.101)

**SIEM/XDR Server:** Production Wazuh Manager (Ubuntu/Linux)



### 2.2 Tools & Systems Used

- **SIEM Platform:** Wazuh (Manager & Agent)
- **Telemetry & Logging:** Windows Sysmon (Event ID 15 Tracking)
- **Threat Intelligence:** VirusTotal API Integration
- **Remediation:** Wazuh Active Response (Python)

### 2.3 Setup & Configuration Steps

- Configured `ossec.conf` on the Windows endpoint to monitor the user download directories using real-time FIM.
- Integrated VirusTotal API via the Manager's configuration to automatically scan unknown FIM hashes.

- Enabled Active Response (`remove-threat.exe`) linked to VirusTotal critical alert Rule ID 87105.
- Deployed custom Sysmon XML configuration and Wazuh local rules (Rule 100085) to capture Mark of the Web (MoTW) URL metadata.

## 2.4 Reconnaissance / Scanning

Not applicable for this specific incident (Direct payload delivery via external download).

## 2.5 Attack Execution

- The malicious payload (`eicar_com.zip` / `eicar.com.txt`) was downloaded directly from an external network into the endpoint, simulating a drive-by download or phishing payload execution.

### 3. Screenshot Evidence / Alerts (SIEM)

| timestamp                  | agent.name    | rule.description                                                                                      | rule.level | rule.id |
|----------------------------|---------------|-------------------------------------------------------------------------------------------------------|------------|---------|
| Sep 1, 2026 @ 18:08:42.788 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:42.761 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:42.749 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:41.715 | Windows_11pro | VirusTotal: Alert - d:\active-response test\eicar.com.txt - 61 engines detected this file             | 12         | 87105   |
| Sep 1, 2026 @ 18:08:40.110 | Windows_11pro | active-response\bin\remove-threat.exe removed threat located at d:\active-response test\eicar.com.txt | 12         | 100092  |
| Sep 1, 2026 @ 18:08:39.951 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:39.941 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:39.941 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:39.867 | Windows_11pro | File deleted.                                                                                         | 7          | 553     |
| Sep 1, 2026 @ 18:08:39.461 | Windows_11pro | VirusTotal: Alert - d:\active-response test\eicar.com.txt - 61 engines detected this file             | 12         | 87105   |
| Sep 1, 2026 @ 18:08:37.210 | Windows_11pro | File added to the system.                                                                             | 5          | 554     |
| Sep 1, 2026 @ 18:08:19.843 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:19.822 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:19.810 | Windows_11pro | Executable dropped in Windows root folder                                                             | 6          | 92217   |
| Sep 1, 2026 @ 18:08:19.289 | Windows_11pro | VirusTotal: Alert - d:\active-response test\eicar.com.txt - 61 engines detected this file             | 12         | 87105   |
| Sep 1, 2026 @ 18:08:17.446 | Windows_11pro | File deleted.                                                                                         | 7          | 553     |

- At 18:08:37, Wazuh FIM detects the creation of eicar.com.txt in the monitored folder (d:\active-response test) under Rule 554. Instantly, the file hash is queried against the VirusTotal API, triggering a critical Level 12 alert (Rule 87105) after 61 security engines flag the file as malicious.

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- Immediately following the VirusTotal detection, the Wazuh Active Response engine triggers the custom script remove-threat.exe at 18:08:40 (Rule 100092). The endpoint confirms the permanent deletion of the threat under Rule 553, completely neutralizing the payload within 3 seconds of introduction.

## 4. Timeline of Detected Events

List detected events in chronological order.

| Date / Time          | Source IP | Destination IP | Event / Alert                             | Notes                                                                          |
|----------------------|-----------|----------------|-------------------------------------------|--------------------------------------------------------------------------------|
| 01/09/2026<br>(Time) | URL       | 192.168.0.101  | <b>Rule 554:</b> File added to the system | FIM detected the new file creation and calculated MD5/SHA256 hashes instantly. |
| 01/09/2026<br>(+1s)  | N/A       | 192.168.0.101  | <b>Rule 87105:</b> VirusTotal Alert       | VT API returned positive malware detection (62+ engines).                      |

## 5. Source of Attack (IP Addresses & Methods Used)

| Attacker Details            | Target Details                                    | Method / Tooling                         |
|-----------------------------|---------------------------------------------------|------------------------------------------|
| External Web Server (EICAR) | Windows 11 Endpoint<br>(d:\active-response test\) | Direct web download / Drive-by execution |

**Note:** Initial attempts to capture the source URL via Sysmon Event ID 15 experienced a race condition due to Wazuh's sub-second FIM read-locks on .crdownload temp files. The threat was remediated before the browser could write the Zone.Identifier metadata.

## 6. Recommendations for Detection & Blocking

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### 6.1 Immediate Containment

- The threat was successfully contained and destroyed via automated Active Response. No manual containment is required.

### 6.2 Longer-Term Recommendations

- **Maintain SOAR Pipelines:** Keep the VirusTotal and Active Response integration enabled for all critical directories (Downloads, Temp, System32).
- **Tune FIM Scans:** Ensure FIM ignores non-executable temp files to reduce API load, while maintaining strict real-time blocks on .exe, .zip, and .dll extensions.
- **Network Blocking:** Feed the identified malicious hash and source domain back into the perimeter firewall to block future download attempts at the network level.

— End of Report —